

- **LINUX COMMANDS**

ls ls -l ls -a	List files of a directory long format listing display all files and hidden
cd [directory] cd ~ cd - cd ..	Change current directory to go to home directory previous directory go to parent directory
pwd	Display the current working directory
mkdir [directory name]	Used to create new directories
touch [filename.filetype]	To create new files
cp [filename] [newfilename]	Copy files on a directory
mv [filename] [newfilename]	Rename a file
mv [filename] ~/directory	Move a file to a new directory
rm [filename] rm -d [foldername] rm -r [foldername]	Deletes files deletes folder, if empty deletes folder and all files inside
echo [message] > [filename]	Insert text to a file
cat [filename]	Display the content of a file

- **GIT COMMANDS**

git init git init [directory]	Initialized a new repo creates a new folder in the path and initialize it as a directory
git status	Show the current state of the directory and staging area
git add [file] git add .	Add a specific file to the commit add all files to the commit
git commit -m "message"	Save code changes with descriptive explanation
git log	Show full commit history for the current branch
git diff	Show unstaged changes compared to the staged changes
git restore [filename] git restore .	Reset the changes on a specific file to the last commit discard all changes for all files
git branch git branch -r git branch -a	List all local branch all remote tracking branch all branch both local and remote
git switch [branch-name] git switch -c [branch-name]	Switch to an existing branch create and switch to a new branch
git merge [branch-name]	Merge the branch into the current branch
git tag git tag -a [tag name] -m "message"	List all tags creates an annotated tag on the current commit with message
git clone [url] git clone [url] [foldername] git clone -b [branchname] [url]	Clone a repo clone into a specific folder clone a specific branch only

- **THREE-TREE ARCHITECTURE** - the three different collections of files used to manage, stage, and track different versions of a project or file.
 1. **Working Directory (Working tree)** - The local workspace. It contains the files you create, modify, or delete.
 2. **Index (Staging Area)** - The preparation zone. It keeps track of the specific changes you want to save.
 3. **Head (Commit History / Repository)** - Tracks the very last commit of the project in the active branch.